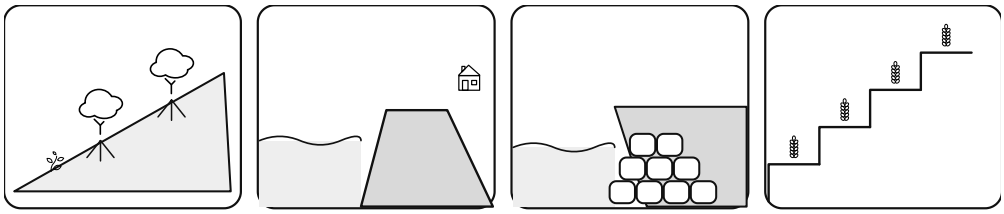
EARTH AND SPACE

Changing Landscapes

Weathering, erosion and deposition by water and wind, solutions to slow erosion, and systems



plant trees and grass | build an embankment | place geo-bags | make terraces

In this pack

- **Learn it:** Wind, water and our changing land ★☆☆
- **Fact check:** Erosion words and solutions ★☆☆
- **Think it through:** Choosing the best solution ★★☆
- **Investigate:** Does a windbreak stop sand blowing away? ★★☆
- **Real-world case study:** Living with a moving river ★★★
- **Review and challenge:** Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · B3 · CHANGING LANDSCAPES

Wind, water and our changing land

Name: Class: Date:

Learning goals

- Explain how wind and water change the shape of the land.
- Compare different ways to slow or stop erosion.
- Explain what a system is, using examples.

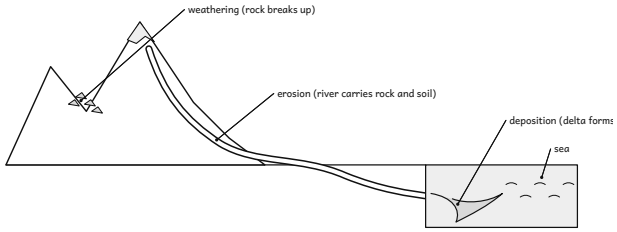
Every year in Bangladesh, rivers eat away their banks and whole villages can fall into the water. At the same time, new islands of sand called **chars** appear in the middle of the rivers. The land is always changing shape. Moving water and wind are the main reasons.

Weathering, erosion and deposition

- **Weathering**: rocks and soil are broken into small pieces by rain, heat, plant roots and ice.
- **Erosion**: the pieces are **carried away** by moving water or wind.
- **Deposition**: when the water or wind slows down, it **drops** the pieces in a new place.

Water changes the land

Fast-flowing rivers carry huge amounts of mud and sand. On the outside of a bend, the water flows fastest and wears away the bank. On the inside of a bend, and where the river meets the sea, the water slows down and drops the mud, building new land. Most of Bangladesh was built over thousands of years from mud carried by the **Padma (Ganges)**, **Jamuna (Brahmaputra)** and **Meghna** rivers. Waves wear away coasts, and heavy rain can wash soil off bare hills and cause **landslides**.



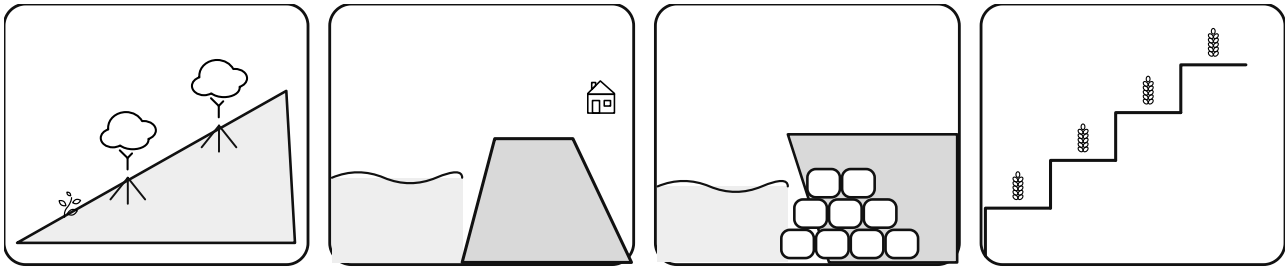
A river wears away land in some places and builds it up in others.

Wind changes the land

Wind picks up dry sand and dust, especially where there are no plants. It can blow sand onto farmland and build **sand dunes** along beaches, like at Cox's Bazar and Kuakata.

Slowing erosion down

People have invented many ways to protect the land. Each has good and bad points.



plant trees and grass | **build an embankment** | **place geo-bags** | **make terraces**

Four ways to slow erosion.

Solution	How it works	Good points	Problems
plant trees and grass	roots hold the soil; leaves break the fall of rain and wind	cheap; helps wildlife	takes years to grow
embankment	a wall of earth or concrete keeps the river or sea back	protects big areas	expensive; can break in floods
geo-bags	big bags of sand placed along a river bank	quick to place	can be washed away
terraces	steps cut into hills slow the water running down	good for farming on hills	hard work to build
windbreaks	rows of trees slow the wind near the ground	protect fields and homes	take space from crops

Thinking in systems

A **system** is a group of parts that work together to do a job that no single part can do alone. A **river** is a system: rain, streams, the main river, its banks and the sea all work together to move water and mud. A **coastal defence** is a system too: a mangrove belt slows the waves, an embankment holds back the water, and **sluice gates** let rain water flow out but stop the sea coming in. If one part fails, the whole system is weaker.

Bangladesh connection

Bangladesh has built thousands of kilometres of embankments around low land near the coast, called **polders**. The Coastal Embankment Project began in the 1960s.

Try this at home

Pour water slowly from a jug onto a pile of soil, and onto a pile of soil covered with grass or leaves. Which one washes away faster?

Key words

weathering	breaking rocks and soil into smaller pieces
erosion	carrying away soil and rock by water or wind
deposition	dropping soil, sand and mud in a new place
char	a sand island that forms in a river
embankment	a long raised bank that holds back water
landslide	soil and rock sliding down a hill
windbreak	a row of trees or a fence that slows the wind
system	parts working together to do a job

FACT CHECK · B3 · CHANGING LANDSCAPES

Erosion words and solutions

★☆☆
Grade 4

Name: Class: Date:

Learning goals

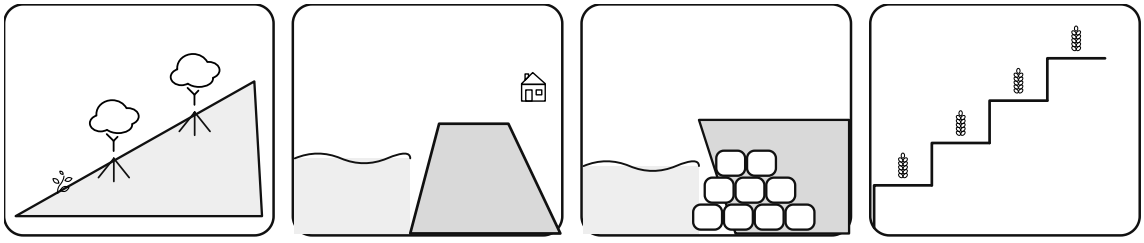
- Use erosion vocabulary.
- Name ways to slow erosion.

1 Match the word to its meaning. [4]

Draw a line to match each pair, or write the letter.

weathering ●	● **A** breaking rocks into small pieces
erosion ●	● **B** dropping mud and sand in a new place
deposition ●	● **C** carrying away soil and rock
landslide ●	● **D** soil and rocks sliding down a hill

2 Name each solution. [4]



A B C D

A	B	C	D

3 Sort the causes. [6]

SORT THESE

a river bank falling in **waves hitting a cliff** **sand blown onto a field**
sand dunes moving on a beach **rain washing soil off a hill**
dust storms in dry land

Water erosion	Wind erosion

4 **True or false?** [3]

Statement	True	False	If false, correct it
1. Plant roots help hold soil in place.			
2. Rivers only take land away; they never build new land.			
3. A system is a group of parts that work together.			

THINK IT THROUGH · B3 · CHANGING LANDSCAPES

Choosing the best solution

★★☆
Grade 4

Name: Class: Date:

Learning goals

- Compare solutions.
- Explain using evidence.

1 **Why is erosion worse on a hill where all the trees have been cut down?** [2]

2 **A village is beside a fast river that is eating away its bank. Compare two solutions and choose the best one. Explain why.** [4]

	Solution 1: _____	Solution 2: _____
good points		
problems		

3 **A coastal defence system has mangroves, an embankment and sluice gates. What would happen if the sluice gates were left open during a storm surge?** [2]

4 **Draw a river bend. Show where erosion happens and where deposition happens.** [2]

River bend

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LEARN · PLAY · WONDER

Does a windbreak stop sand blowing away?

★★☆ Grade 4

Name: Class: Date:

Learning goals

- Carry out a fair test.
- Measure mass.
- Evaluate a solution.

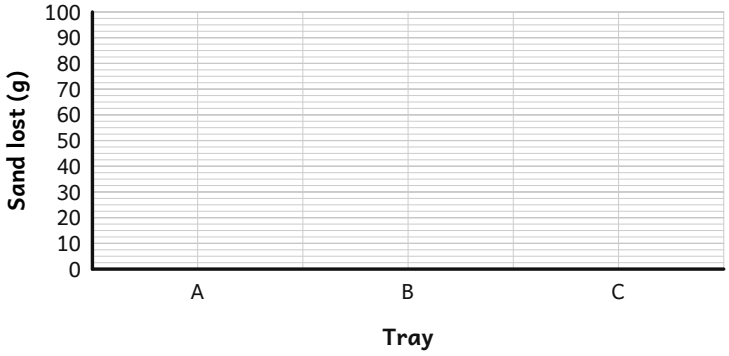
Class 4B put 200 g of dry sand in each of three trays. They blew air across each tray with a fan for 1 minute, then weighed the sand that was left. Tray A had nothing, tray B had a row of small sticks (a model windbreak), and tray C had a row of sticks and the sand was damp.

1 **Complete the plan.** [3]

Change	
Measure	
Keep the same	

2 **Complete the table and draw a bar chart of the sand lost.** [4]

Tray	Sand at start (g)	Sand left (g)	Sand lost (g)
A: nothing	200	110	
B: windbreak	200	170	
C: windbreak + damp	200	195	



3 **Write a conclusion.** [2]

4 **How could farmers use these results?** [1]

REAL-WORLD CASE STUDY · B3 · CHANGING LANDSCAPES

Living with a moving river

★★★ Grade 4

Name: **Class:** **Date:**

Learning goals

- Understand river erosion in Bangladesh.
- Use data about land loss.
- Compare solutions from different viewpoints.

Science · Geography · Maths · Economics · Citizenship

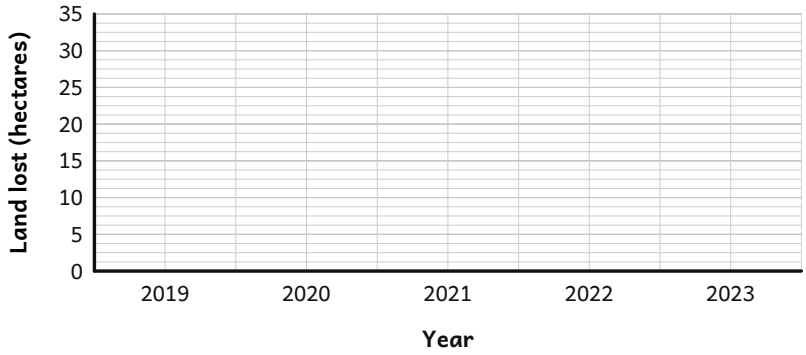
The **Jamuna** and **Padma** are among the biggest rivers in the world. Each monsoon they carry enormous amounts of water and sand. Their banks can collapse suddenly, and thousands of families lose their homes and fields to the river every year. People who lose their land often move to new chars or to the cities.

Engineers protect important places, such as towns and the Padma Bridge, with **concrete blocks**, **geo-bags** and walls along the banks. This is called **river training**. But it is expensive, and protecting one place can make the river push harder somewhere else.

Based on reports about riverbank erosion in Bangladesh. Data below are example values.

1 **Land lost to the river at one village (example data).** [4]

Year	2019	2020	2021	2022	2023
Land lost (hectares)	12	30	8	22	10



a) Draw the bar chart.

b) What is the total land lost in five years?

c) The worst years had very big monsoon floods. Why does this make sense?

2 **What might each person want?** [3]

Person	View
a farmer on the bank	
an engineer	
a family on a new char	

3 **If you were in charge of the money, how would you protect the village? Explain your plan.** [2]

REVIEW AND CHALLENGE · B3 · CHANGING LANDSCAPES

Show what you know

★★★
Grade 4

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

T	I	M	K	L	W	I	N	D	G
V	G	H	T	G	I	L	D	N	T
R	N	K	E	K	A	B	K	A	L
I	B	C	H	A	R	O	O	T	S
V	S	C	E	R	O	S	I	O	N
E	Y	A	R	S	O	I	L	O	O
R	W	U	N	A	M	O	L	B	I
V	P	G	O	D	D	H	L	P	I
F	T	S	Y	S	T	E	M	W	E
Y	U	U	B	B	A	N	K	J	J

bank char erosion river roots sand soil system wind

2 Which is the best way to stop soil washing off a bare hill? [1]

- ◯ cut down more trees
- ◯ plant trees and grass
- ◯ build a road
- ◯ burn the grass

3 Where does a river drop most of its mud? [1]

- ◯ where it flows fastest
- ◯ where it slows down
- ◯ at the top of a waterfall
- ◯ on the outside of a bend

4 **Draw a coastal defence system with three parts. Label each part and its job.** [3]

Coastal defence system

5 **Challenge. Why might protecting one part of a river bank make erosion worse further down the river?** [2]

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
explain weathering, erosion and deposition	○	○	○
describe how rivers and wind change the land	○	○	○
compare ways to slow erosion	○	○	○
test a model windbreak	○	○	○
explain what a system is	○	○	○
suggest how to protect a village from river erosion	○	○	○